Pricing and Risk of Structured Products: Greeks, Monte Carlo and Hedging

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1 The Core Idea

A structured product is a bond plus options, and its risk is the risk of those options. A Phoenix autocall, for example, pays coupons, can be redeemed early, and loses capital if the underlying ends low. It has no closed-form price. A risk manager still needs its price, its sensitivities, and the effect of hedging it.

The repository (`structrisk`) does four things for this.

- **Closed-form Greeks** of a vanilla option, from first order (delta, vega) to third order (speed, zomma, color).
- **Monte Carlo pricing** of a Phoenix autocall and of other notes, with confidence intervals. Sensitivities come from bump-and-revalue with common random numbers.
- **Portfolio risk:** Greeks added across positions, in euros, with a spot-by-volatility stress grid.
- **Hedging simulation:** the replication error of a delta-hedged option, with zero costs or with some.

2 Why It Matters

- **The coupon is a price.** With a 4% coupon per half year the Phoenix is worth 100.71 per 100 of notional, so at par the buyer gets 0.71 more than fair value. The coupon that makes the price 100 is 3.63%. Every 1 point of coupon moves the price by 1.91.
- **The holder is short volatility.** The price falls from 104.74 at 15% volatility to 95.27 at 35%, about 0.5 per vol point. The investor has sold the capital protection, and a rise in volatility makes that put more valuable.
- **Gamma lives in short maturities.** The ATM gamma at one month is 5.3 times the gamma at two years, so a short-dated position needs far more hedging for the same size.
- **Some Greeks cannot be measured.** The Phoenix payoff has jumps at the barriers. With 200,000 paths the vega is stable (about -52) but the gamma changes sign from one seed to the next. A report that shows a gamma for such a product shows noise.

3 Assumptions

Each line states what the code assumes and which way the result moves if it is wrong.

- **Geometric Brownian motion, flat volatility.** The autocall uses 25% volatility, $r = 3\%$, $q = 2\%$ and no skew. Real surfaces put more volatility on low strikes, which raises the probability of crossing the 60% barrier, so a flat volatility overstates the price of a product that is short that put.
- **Barriers on observation dates only.** The autocall and coupon barriers are tested on four semi-annual dates, and the 60% capital barrier only at maturity. A continuous capital barrier would be breached more often and the price would be lower.
- **Exact simulation on the dates.** Spot is drawn at the observation dates with the exact lognormal step (no time discretisation error), with 200,000 antithetic paths and seed 2026.
- **Memory coupon.** Missed coupons accumulate and are paid in full when the coupon barrier is next met. The last date pays the coupon and the redemption.
- **Risk-free discounting only.** Cash flows are discounted at 3%. There is no issuer spread, no funding cost and no margin, so the price is not what a bank would quote. An issuer spread would lower the value of every promised payment.
- **Bump-and-revalue Greeks.** Spot is bumped by 1%, volatility by 0.01, with the same seed in every revaluation and the barrier levels fixed at the initial spot. The result is noisy for jump payoffs and the bump size matters.
- **Hedging experiment.** The hedge uses the same volatility as the simulated paths (unless a realised volatility is given), rebalances on a regular grid, and charges costs in bp of the traded notional. Dividends are received on the shares held.
- **Portfolio.** One underlying, static positions, Greeks per unit times quantity. The stress grid is a second-order expansion in spot with a vanna term, not a repricing. In the example, the note and the bonus certificate are bumped with strike, barrier and reference spot fixed at 100, so only the spot moves. The certificate gamma comes from a 100,000-path simulation and is noisy (see Results).

4 Method

4.1 Black-Scholes-Merton Greeks

With dividend yield q, $d_1 = \dfrac{\ln(S/K) + (r - q + \frac{1}{2}\sigma^2)T}{\sigma\sqrt{T}}$ and $d_2 = d_1 - \sigma\sqrt{T}$, the call Greeks are:

Greek	Formula	Note
Delta	$e^{-qT}N(d_1)$	per unit of spot
Gamma	$\dfrac{e^{-qT}\varphi(d_1)}{S\sigma\sqrt{T}}$	same for call and put
Vega	$Se^{-qT}\varphi(d_1)\sqrt{T}$	per 1.00 of vol
Vanna	$-e^{-qT}\varphi(d_1)\, d_2/\sigma$	$\partial\Delta/\partial\sigma$
Volga	$\text{Vega} \cdot d_1 d_2/\sigma$	$\partial\text{Vega}/\partial\sigma$

Charm, speed, zomma, color and veta are also implemented in closed form. Theta and rho are returned per year and per 1.00 of rate. Delta says how much stock to hold, gamma how often to change it, vega how much the price depends on volatility, and the cross terms (vanna, volga) how those exposures move when spot or volatility moves.

4.2 Phoenix autocall by Monte Carlo

Spot paths follow geometric Brownian motion with drift $r - q$, simulated at the observation dates with antithetic variates. At each date t_j with performance $P_j = S_{t_j}/S_0$:

- **Early redemption:** if $P_j \geq$ the autocall barrier (and j is not the last date), the notional is repaid at t_j.
- **Coupon with memory:** if $P_j \geq$ the coupon barrier, the holder receives the coupon times the number of periods since the last payment. Missed coupons are paid when the barrier is next met.
- **At maturity:** if the product was never called, the notional is repaid when $P_n \geq$ the protection barrier, and $N \times P_n$ otherwise.

The price is the mean of discounted cash flows, $\sum_j c_j e^{-rt_j} + R\,e^{-r\tau}$, with a 95% interval $\pm 1.96\, s/\sqrt{n}$. The product parameters are the coupon (4% per period), the barriers (100%, 70%, 60%) and the four dates. The price rises with the coupon and falls as the coupon barrier or the protection barrier rises.

4.3 Greeks of the structured products

Greeks are central differences of the price with the same seed in every bump (common random numbers), so the payoff discontinuities do not turn into noise. Delta uses a spot bump of 1, vega a volatility bump of 0.01, gamma the three-point second difference. The library also has pathwise and likelihood-ratio delta estimators for a vanilla call, which serve as a check of the bump method.

4.4 Portfolio aggregation

For positions with quantity q_i, multiplier m_i and spot S:

$$\text{Delta-equivalent exposure} = \sum_i q_i m_i\, \Delta_i\, S,$$

$$\text{Gamma P\&L for } +1\% = \tfrac{1}{2} \sum_i q_i m_i\, \Gamma_i\, (0.01\, S)^2,$$

$$\text{Vega P\&L per vol point} = 0.01 \sum_i q_i m_i\, \mathcal{V}_i.$$

The stress grid uses, for each position, $\Delta\, dS + \tfrac{1}{2}\Gamma\, dS^2 + \mathcal{V}\, d\sigma + \text{Vanna}\; dS\, d\sigma$, with $dS = \text{shock} \times S$.

4.5 Delta-hedging simulation

A call is sold for its Black-Scholes premium. At each of n dates, the position holds the model delta in shares, pays a cost of c bp on the traded notional, and earns the dividend on the shares. At expiry, the replication error is the portfolio value minus the option payoff.

5 Method Comparison

Method	Strength	Weakness
Closed-form Greeks	Exact, instant, any order	Vanilla payoffs only
Bump-and-revalue	Works for any payoff	Noisy for jumps; bump-size choice
Pathwise delta	Low variance for continuous payoffs	Fails when the payoff jumps
Likelihood ratio	Works for jumps	Standard deviation 6 times higher on a call
Control variate (price)	Standard error 2.3 times lower on a call	Needs a known mean

On a vanilla call with 200,000 paths, the standard deviation of the delta across 20 seeds is 0.0004 for the pathwise and bump estimators and 0.0023 for the likelihood ratio (closed form 0.5506). The control variate cuts the price standard error from 0.0327 to 0.0143. Neither shortcut applies to the Phoenix, so the bump with common random numbers is the working method, and its noise has to be read next to each Greek.

6 Results

Greeks. ATM call, 1 year, $r = 3\%$, $q = 2\%$, $\sigma = 22\%$: price 9.0394, delta 0.5506, gamma 0.01756, vega 38.635, theta -0.0124 per day. The put has delta -0.4296 and the same gamma and vega. Second-order terms: vanna 0.1134, volga -1.762, which agree with a bump of the closed-form vega to better than 10^{-6}. ATM gamma at 1 month (0.08 year) is 0.06395 against 0.01203 at 2 years, a ratio of 5.32.

Maturity (years)	Delta	Gamma	Vega	Theta per day
0.08	0.5167	0.06395	11.25	-0.0436
0.25	0.5283	0.03598	19.79	-0.0249
1.0	0.5506	0.01756	38.63	-0.0124
2.0	0.5640	0.01203	52.91	-0.0085

Gamma falls with maturity roughly as $1/\sqrt{T}$ (the ratio would be 5.0 for exactly $1/\sqrt{T}$), and vega rises as $\sqrt{T}$. A short option has high gamma and high theta, and a long one has high vega. The delta is above 0.5 because the drift $r - q$ is positive. The table is for one strike: away from the money gamma and vega fall quickly.

Phoenix autocall. Spot 100, volatility 25%, $r = 3\%$, $q = 2\%$, observations every six months for two years, 4% coupon per period, barriers 100% (call), 70% (coupon) and 60% (capital), 200,000 paths.

Quantity	Value
Price (% of notional)	100.71 (95% interval 100.64 to 100.77)
Called at 0.5 year	47.59%
Called at 1.0 year	11.93%
Called at 1.5 years	6.10%
Never called	34.38%
Capital loss, given never called	22.71%

Raising the call barrier from 95% to 105% moves the price from 100.51 to 101.05. Recomputed with the repository function (same seed), the price has delta 0.173 per unit of spot and vega

−52.7 per 1.00 of volatility, so the holder is short volatility. Almost half of the paths are called at the first date and the investor then holds a six-month bond with one coupon. The tail is in the 34% never called: 22.7% of those, or 7.8% of all paths, end below 60% and lose capital. The table does not show the distribution of the loss, only its probability.

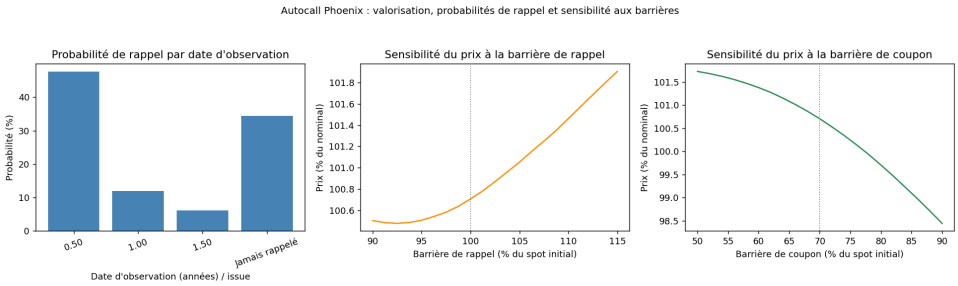


Figure 1: Phoenix autocall: probability of each outcome and price against the call and coupon barriers (labels in French).

Portfolio. Long 1,000 calls (strike 105) and short 600 puts (strike 95), both 1 year, plus 50 capital-protected notes and 30 capped bonus certificates, on one underlying (spot 100, volatility 22%). The spot-by-volatility grid (spot ±20%, volatility ±10 points) goes from EUR −18,482 to +24,661.

Position	Value (EUR)	Delta eq. (EUR)	Gamma, +1% (EUR)	Vega per pt (EUR)
1,000 calls, 105	6,952	46,418	+8.87	+390.18
−600 puts, 95	−3,442	20,513	−4.94	−217.56
50 protected notes	51,529	16,146	+1.44	+63.26
30 bonus certificates	30,966	18,034	−0.18	−134.20
Total	86,006	101,111	+5.18	+101.68

The delta-equivalent exposure of EUR 101,111 is 118% of the market value. The two vanilla options give 66,931 of it (the short put adds delta, since a short put is long the market) and the two structured products give 34,180. The note has delta 3.23 per unit: it holds 80% of a call spread on 10 shares of notional, and the call spread has a delta of about 0.40. The certificate has delta 6.01 per unit, about 0.60 of its 10 shares of notional. The vega goes the other way for the two products. The note is long volatility (+63.26 EUR per point) because it owns a call spread, and the certificate is short volatility (−134.20) because the holder has sold the 70% barrier put. Together they pull the portfolio vega from EUR +172.62 for the two options down to +101.68. The certificate Greeks come from 100,000 paths with the same seed in every bump. With seeds 1, 2 and 3 its delta is 5.93, 6.07 and 5.92 and its vega −453, −452 and −444, but its gamma is −0.44, −0.25 and −0.14 per unit against −0.012 in the table, so its gamma line (EUR −0.18) is noise.

Delta hedging. Short call, spot 100, strike 100, 6 months, volatility 25%, 40,000 paths, premium 7.2054. The error is in the currency units of the spot. The last column is the Derman–Kamal leading-order formula for the call, $\sqrt{\pi/4}\,\mathcal{V}\sigma/\sqrt{n}$ with vega 27.74 per 1.00, added for comparison (it is not in the code).

Rehedges	Std, no cost	Formula	Mean, 10 bp	Std, 10 bp
1	5.3962	6.146	−0.1120	5.3962
4	2.8543	3.073	−0.1235	2.8641
12	1.6970	1.774	−0.1660	1.7075
52	0.8388	0.852	−0.2812	0.8523
252	0.3894	0.387	−0.5570	0.4407

More rehedging narrows the error but costs more. The mean cost grows from 0.11 to 0.56 between 1 and 252 rehedges. About 0.055 of the first figure is the opening hedge (10 bp on 0.55 shares at 100), and from 52 to 252 rehedges the cost doubles (0.28 to 0.56) while the number of trades is 4.8 times larger. With one rehedge the error (5.40) is 75% of the premium. The mean without costs is within 0.06 of zero at every frequency (the standard error is 0.027 for one rehedge and 0.002 for 252). The formula is within 2% of the simulation from 26 rehedges and 14% above it at one, where the leading-order result does not apply.

7 Worked Example

Greeks by hand for the ATM call above ($S = K = 100$, $T = 1$, $r = 3\%$, $q = 2\%$, $\sigma = 22\%$):

$$\begin{aligned} d_1 &= \frac{(0.03 - 0.02 + 0.5 \times 0.22^2) \times 1}{0.22} = \frac{0.0342}{0.22} = 0.1555, \\ N(d_1) &= 0.5618, \quad \varphi(d_1) = 0.3942, \quad e^{-qT} = 0.9802, \\ \Delta &= 0.9802 \times 0.5618 = 0.5506, \\ \Gamma &= \frac{0.9802 \times 0.3942}{100 \times 0.22} = 0.01756, \\ \mathcal{V} &= 100 \times 0.9802 \times 0.3942 = 38.63. \end{aligned}$$

The price follows from $N(d_2) = N(-0.0645) = 0.4743$: $C = 98.02 \times 0.5618 - 97.04 \times 0.4743 = 9.04$. The Black-Scholes equation holds with the code's Greeks: $\Theta + \frac{1}{2}\sigma^2 S^2 \Gamma + (r-q)S\Delta - rC = -4.5293 + 4.2498 + 0.5506 - 0.2712 = 0.0000$, with theta per year. Theta pays for gamma, apart from the carry.

One Phoenix path (illustrative): performances at the four dates are 65%, 80%, 72%, 58%, with notional 100. The coupon is $4\% \times 100 = 4$ per period.

- $t = 0.5$: 65% is below the 70% coupon barrier, no coupon (one period is owed).
- $t = 1.0$: 80% is above 70%, so the coupon is paid for two periods: $4 \times 2 = 8$.
- $t = 1.5$: 72% is above 70%, coupon 4.
- $t = 2.0$: 58% is below 70%, no coupon. It is also below the 60% protection barrier, so redemption is $100 \times 0.58 = 58$.

Present value at $r = 3\%$: $8e^{-0.03} + 4e^{-0.045} + 58e^{-0.06} = 7.764 + 3.824 + 54.622 = 66.21$. The same path with a performance of 105% at the first date pays 4 and redeems 100 at $t = 0.5$.

Capital-protected note by decomposition (notional 1,000, protection 100%, participation 80%, strike 100%, cap 130%): the zero coupon is $1000\, e^{-0.03} = 970.45$. The call spread is $C(100) - C(130) = 9.0394 - 1.5221 = 7.5173$, and the option leg is $0.8 \times (1000/100) \times 7.5173 = 60.14$. The note is worth $970.45 + 60.14 = 1030.58$. The 3.1% above par is the value of the participation. A real issue would also take out the issuer's margin and funding spread.

Sensitivity. All lines rerun the repository code.

- *Volatility.* The Phoenix is worth 104.74, 103.15, 100.71, 97.99 and 95.27 at 15, 20, 25, 30 and 35%. The probability of a capital loss given no call goes from 2.4% to 43.7% over the same range.
- *Barriers.* With a capital barrier of 50, 60 and 70% the price is 102.76, 100.71 and 98.48. With a coupon barrier of 50, 60, 70, 80 and 90% it is 101.73, 101.38, 100.71, 99.71 and 98.44.

- *Rates and dividends.* With $r = 0, 3, 5\%$ the price is 102.77, 100.71 and 99.33. With $q = 0, 2, 4\%$ it is 101.42, 100.71 and 99.85. A higher rate cuts the value of the notional redeemed late.
- *Monte Carlo noise.* The price is 100.68 (standard error 0.140), 100.73 (0.063), 100.71 (0.031) and 100.72 (0.016) with 10,000, 50,000, 200,000 and 800,000 paths. Seeds 1 to 5 at 200,000 paths give 100.77, 100.72, 100.70, 100.75 and 100.72.
- *Greeks across seeds.* At 200,000 paths, seeds 1 to 3 give delta 0.161, 0.173, 0.171, vega -50.8, -52.3, -52.8 and gamma -0.010, -0.019, -0.023. With seed 2026 and spot bumps of 0.5, 1 and 5%, gamma is $+0.019$, $+0.0003$ and -0.014. Delta and vega are usable to about two significant figures and gamma is not usable.
- *Realised volatility in the hedge.* The call is sold at 25% and hedged 52 times, while the paths have a volatility of 15, 20, 25 and 30%. The mean error is $+2.83$, $+1.41$, $+0.00$ and -1.41 against price gaps of $+2.78$, $+1.39$, 0 and -1.39. The standard deviation is 0.94, 0.82, 0.84 and 1.16.

8 Limits and Extensions

- **Constant volatility:** no stochastic volatility or jumps. The volatility surface in the examples is a synthetic parametrisation, not market data.
- **No credit risk:** issuer risk is not in the structured product prices, although it is a large part of the risk of a note.
- **Barrier monitoring:** barriers are checked on the simulation grid, not continuously.
- **Portfolio example:** the certificate gamma is noise (see Results), and the stress grid is a second-order expansion with a vanna term, not a full revaluation: for the two vanilla legs it is within about 5% of repricing at the corners (14,956 against 15,006 at +20% spot, 18,031 against 17,119 at +20% spot and +10 vol points).
- **Hedging scaling:** the error does not fall exactly as $1/\sqrt{n}$. From 1 to 252 rehedges, the rule gives $5.3962/\sqrt{252} = 0.340$ and the simulation gives 0.389. These are model outputs, not recommendations.

Three extensions would address these limits. A local or stochastic volatility model, calibrated on a skewed surface, would price the autocall with the skew that a real desk sees. A repricing grid (the structured products revalued at each shocked point) would replace the Taylor expansion.

9 Conclusion

The repository gets the vanilla Greeks exactly and the Phoenix price to within 0.03, which is the easy part. What a risk manager should take from it is that the sensitivities of a jump payoff come with a noise level, and that the way a product is bumped can change the answer: bumped with its strike reset to the new spot, the note and the certificate showed no delta, a third of this portfolio's exposure.